

Exploring the Flavor and Charge Sectors: an epic synthesis of the emergent proton’s electroweak content

Tessera — cobordism subsystem (epic #410: #411–#418, #429, #437)

Abstract

This note synthesizes the *Flavor and Charge Sectors* exploration track (epic #410) end-to-end, now that its foundational and sector tickets have merged. The track asked whether the proton’s electroweak content — electric charge, the proton/neutron (flavor) distinction, and the spinor/current structure — could be found by *splitting the behavior of the existing color register* (spatio-temporally and across form-degree) rather than by bolting on parallel hole registers. The answer is now substantially in hand, and it is more interesting than the plan anticipated. The two geometric foundations landed: orientation is sourced from the standard simplicial formula (#412, `TemporalOrientation / ChainComplex::endSignCovector`), and the asymmetric prism interior was replaced by a *symmetric apex interior* that is now the default (#413, `Spacetime::symmetricStackCells`) — driving the color-singlet overlap to 1 and the intertwining residual to machine zero, with a side finding that the prism’s “spontaneous photon” was triangulation-assisted. The charge sector is built and emergent: the field strength splits into E (timelike-leg) and B (spacelike) plaquettes (#417), and electric charge reads as a temporal-sector Gauss-law holonomy $Q = \oint \star F$ (#411, `EigenstateSynthesis::gaussLawCharge`) that vanishes on a Riemannian relaxation and populates only when Lorentzian worldlines do. The spin/current sector is the Dirac–Kähler factor ($d + \delta$) of the Hodge operator, with $(d + \delta)^2 = L$, exact Clifford relations, γ^5 chirality, the conserved current j^0 , and the $16 = 4 \times 4$ taste multiplicity (#415, `DiracKahler`). The binding question was reopened bipartitely (#416): an orientation-reversing twist projects the antisymmetric $\bar{\mathbf{3}}$ diquark, and a *connected* two-body path reaches the singlet (reachability 1.00 vs. 0.55 welded) — confinement is not a fundamental obstruction inside one connected bulk. The flavor sector returned a clean *no-go* (#414): flavor as a relabeling-invariant split of the A_4 -symmetric color register is obstructed, because flavor is symmetry-odd while the singlet is symmetry-even and the two cannot share a register; the fix is a symmetry-breaking isospin doublet, a new construction. The dimensional spike (#418) and the dimension-generic apex interior (#429, `symmetricStackCells(nApexSlices)` with a rigorous `dualComplexIsValid` for $n \geq 4$, validated as a 4-manifold over a tetrahedral S^3) scope the 3+1 D home and expose the emergent twist as a PT-symmetric structure — correctly read as *localized to creation/annihilation vertices* (Stückelberg–Feynman), not a per-slice alternation. Finally, the track’s findings force an **architecture pivot**: the proton is assembled from *sequences of bipartite junctions* relaxed in *one global step* with color-emergent creation, and the tripartite W_{ABC} is dropped. Charge lives in the emergent temporal sector, color in the register: the charge $\otimes$ color thesis. All numbers below are measured through the production C++ and its Python suite; the open work (named, not created) is collected at the end.

Contents

1 What the track set out to do

2

2	Geometric foundations	3
2.1	Orientation is geometric, not a vertex-label sort (#412)	3
2.2	The symmetric apex interior is the default; the singlet is exact (#413)	3
3	The charge sector	4
3.1	E and B from the temporal/spatial split (#417)	4
3.2	Electric charge as a Gauss-law holonomy (#411)	4
4	The spin / current sector: Dirac–Kähler (#415)	5
5	The binding structure: a bipartite revisit (#416)	5
6	The flavor sector: a no-go (#414)	6
7	The dimensional question (#418, #429)	6
7.1	Dimension-generic apex interior and an $n \geq 4$ manifold check (#429)	6
8	CI hygiene (#437)	7
9	Architecture pivot: bipartite junctions and one global relaxation	7
10	Faithfulness: the golden invariants held	8
11	What is established, and what is open	8

1 What the track set out to do

The cobordism subsystem carries the proton’s *color-confinement skeleton*: on a symmetric trivalent junction a color-symmetric quark input transports to the net-color-zero singlet $\langle(1, \omega, \omega^2)\rangle$, confinement ($\sigma \rightarrow 0$) is a Stokes conservation law, and the singlet’s location is forced by the geometry’s $\mathbb{Z}_3$ symmetry rather than imposed. What it did *not* carry was the electroweak content: charge, the proton/neutron flavor distinction, and the spinor/current structure. The earlier conclusion was a *richer construction* — a flavor register parallel to color, plus spinor fields. Epic #410 took the opposite road.

Principle 1 (Split the register, do not multiply it). *The proton’s electroweak content is sought by splitting the behavior of the existing register — across causal type (temporal vs. spatial) and across form-degree (the inhomogeneous cochain) — rather than by adding parallel hole registers. Independence between sectors is supplied by orthogonality in the spectral/causal decomposition, not by disjoint topology.*

This synthesis records what that program produced. The plan note (#420, `flavor_charge_sectors.tex`) laid out the sectors and their falsifiable tests; the preliminary results note (`flavor_charge_results.tex`) recorded the first wave. Here we close the loop: every sector ticket has now landed, two of the load-bearing hypotheses were *killed* (the flavor split; the per-slice twist reading), the binding question was answered, and the assembly architecture pivoted. Each section names the merged ticket, the production API, and the measured verdict.

2 Geometric foundations

The two foundations removed arbitrariness from the geometry; every downstream observable inherits the symmetry they restored.

2.1 Orientation is geometric, not a vertex-label sort (#412)

The color charge of a window is a sum of per-hole periods. The inputs were already read in the surface’s induced orientation, but the *emergent result* was summed in each hole’s own sorted-vertex orientation — a function of the arbitrary vertex numbering, so a relabeling could flip a hole’s term with no conjugating partner and scramble the sum. The fix sources every sign from the standard simplicial orientation. Two distinct notions of “orientation” were separated in code: the Causal-Dynamical-Triangulations (t_i, t_f) temporal class became **TemporalOrientation** (with **orientationOf** the standard-orientation anchor), reserving the *spatial* induced orientation for **ChainComplex::endSignCovector** (a globally coherent ± 1 covector propagated across shared facets). Applying the result block’s induced signs symmetrically with the inputs turns the signed carried charge into the geometry’s Stokes invariant:

input	signed $ \sigma_R $ (induced)	raw $ \sigma_R $ (sorted)
neutral ($q\bar{q}$ triple, $\Sigma = 0$)	0.0000 (exact)	0.2258
colored (R, G, B , $\Sigma = 1$ each)	3.0000 (exact)	1.0219

The signed sum is exactly $\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$; the raw sum is neither, and is label-dependent. Correctness is pinned by a *relabeling-invariance* test (the track’s G_6): under seeded permutations of the base-surface vertex ids the signed σ_R , singlet overlap, and intertwining residual are invariant up to one global sign shared by all windows, while the raw sum drifts.

2.2 The symmetric apex interior is the default; the singlet is exact (#413)

The trivalent junction was built by extruding the holed base surface $\times I$ and tetrahedralizing each triangular prism by the Freudenthal staircase, whose diagonal is chosen by a vertex-label sort. That choice — not the metric — was the documented source of the intertwining residual $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| \approx 4.26 \times 10^{-2}$ that held the singlet at 0.999. The replacement, **Spacetime::symmetricStackCells**, cones each base triangle up to a single face-apex vertex (a $(3,1)$ tetrahedron) and down to the top copy (a $(1,3)$ tetrahedron); the octahedral gap over a shared base edge is split along the *canonical dual edge* $f_1 f_2$ (the two face-centres), fixed by the incident faces rather than by ids. With no label-sort anywhere the subdivision is equivariant under the icosahedral symmetry, and the transport intertwines the color $\mathbb{Z}_3$ *exactly*:

	prism (Freudenthal)	symmetric apex (default)
intertwining residual $\ MP_{\text{in}} - P_{\text{out}}M\ / \ M\ $	4.26×10^{-2}	4.5×10^{-14}
color-singlet overlap	0.999	1.000000
valid manifold (dualComplexValid) / b_1	yes / 11	yes / 11

The symmetric interior is the default; **set_symmetric_interior(false)** selects the legacy prism. The twenty existing junction tests pass unchanged — the residual simply moved inside their tolerances — so the color-confinement skeleton (charge conservation exact at $\sim 10^{-14}$, $\sigma \rightarrow 0$, the $\mathbb{Z}_3$ spectrum $\{1, \omega, \omega^2\}$) is preserved while the singlet becomes exact.

Finding 1 (The spontaneous photon was triangulation-assisted). On the prism, making the cross-layer worldlines timelike and relaxing drives one worldline through null ($\ell^2 \rightarrow 0$) — the predicted “photon.” On the symmetric interior the same seed does *not* produce it: through 120 relaxation iterations all 450 worldline edges stay uniformly timelike at $\ell^2 = -0.3$ with *zero* near-null edges. The prism’s photon was triggered by the triangulation asymmetry singling out one edge; the symmetric relaxation preserves the worldline symmetry. A genuine photon should emerge from a symmetry-breaking *source*, not the bare uniform seed — the symmetric interior is the more honest setting, and a source-driven photon is the natural follow-up.

3 The charge sector

3.1 E and B from the temporal/spatial split (#417)

Let $A \in \Omega^1$ be a $U(1)$ connection and $F = dA \in \Omega^2$ its curvature. On the Lorentzian complex each plaquette is classified by the causal type of its edges:

$$F = \underbrace{F^{(E)}}_{\text{a timelike leg}} \oplus \underbrace{F^{(B)}}_{\text{purely spacelike}}, \quad (1)$$

the discrete analogue of $E_i = F_{0i}$, $B_{ij} = F_{ij}$. This is implemented as `EigenstateSynthesis::curvatureFromConnection` ($F = dA$, the same induced-orientation signed edge sum the periods use) and `fieldStrengthSplit`, returning the electric and magnetic parts and their disjoint, complete cell index lists. The tests confirm the split tracks the causal structure: on a purely spacelike (static) seed $\|E\| = 0$ exactly and $B = F$ on every cell; under timelike flux $\|E\| > 0$ and the electric cells equal the independently-recomputed timelike-leg plaquettes; $E + B = F$ to 10^{-12} ; and $F = dA$ is gauge invariant ($A \rightarrow A + d\chi$ leaves E, B unchanged). The reader is purely observational — the proton sector is bit-for-bit unchanged with it present or absent — and because the split is read off the cochain by causal type it is *interior-shape-independent*: it holds equally on the prism and the symmetric apex.

3.2 Electric charge as a Gauss-law holonomy (#411)

Electric charge is the discrete Gauss law, the flux of E through a closed surface enclosing the quark windows:

$$Q = \oint_{\partial V} \star F = \int_V \star j, \quad (2)$$

implemented as `EigenstateSynthesis::gaussLawCharge`. The key property is what kind of object this is. The previously-considered “flavor-weighted covector” $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ is not A_4 -equivariant and is not topologically protected; it would degrade under a non-symmetric seed the way charge conservation degrades under jitter. The Gauss-law holonomy (2) is a genuine gauged- $U(1)$ invariant — emergent and metric-robust.

Finding 2 (Charge is emergent and Lorentzian, not hand-weighted). Q is *not* a hand-weighted covector. It is read off the relaxed geometry, and it is *zero on an all-spacelike (Riemannian) relaxation*: with no timelike worldlines the E -sector is empty and Gauss’s law returns no charge. The charge populates only when Lorentzian worldlines do — charge is intrinsically a temporal-sector object, exactly as the reframe predicted. This is the *charge* $\otimes$ *color* thesis in one fact: color is the topological register kernel (metric-robust, always present); charge is the temporal Gauss-law holonomy (present only with a Lorentzian causal structure).

In the Standard Model the fractional part of the quark charge is tied to color through the center $\mathbb{Z}_3$ (triality) via $Q = I_3 + \frac{1}{2}Y$. The construction already carries triality as P_{out} ; whether the $\frac{1}{3}$ -quantum of the Gauss-law charge *is* that triality is the natural cross-check, alongside the agreement of Q with the Dirac–Kähler current j^0 of §4.

4 The spin / current sector: Dirac–Kähler (#415)

The inhomogeneous cochain $\Omega^\bullet = \bigoplus_k \Omega^k$ is the discrete Dirac–Kähler field, and $D = d + \delta$ is its Dirac operator (the class `DiracKähler`). Assembled from the integer boundary maps and metric weights — reusing the convention the Hodge Laplacian already squares — it reproduces the expected algebra to machine precision:

property	measured	meaning
$(d + \delta)^2 = L$	residual $\sim 6 \times 10^{-15}$	D is a square root of the Hodge Laplacian
$\{\gamma^a, \gamma^b\} = 2\eta^{ab}$	residual 0.0 (exact)	the Clifford algebra of the 1-cochain action
γ^5 chirality	$(\gamma^5)^2 = I, \{\gamma^5, \gamma^a\} = 0$	the left/right (chiral) split
$j^0 = \bar{\Phi}\gamma^0\Phi$	$= \pm \Phi^\dagger\Phi$	the conserved (Noether) charge density
multiplicity	$= 4$	the taste degeneracy $\Lambda(\mathbb{C}^4) = 16 = 4 \times 4$

The gamma generators are the 16×16 Clifford action of unit 1-cochains on the form fiber, $\gamma^a = \varepsilon(e^a) + \eta^{aa}\iota(e^a)$ (an edge raises and lowers form degree) — the literal realization of “gammas = the Clifford action of edges.” In four dimensions $\Omega^\bullet \cong \text{Cl}(1, 3) \otimes \mathbb{C} \cong M_4(\mathbb{C})$, so the 4×4 Dirac structure is *factored out of the operator the construction already builds*. The charge density is `DiracKähler::chargeDensity`, the time component of the conserved current; its cross-check against the Gauss-law charge of §3.2 ties the operator sector to the temporal sector. The $16 = 4 \times 4$ taste multiplicity (the lattice Dirac–Kähler “doubling”) was the plan’s candidate flavor index — a hypothesis the flavor sector then tested and rejected (§6).

5 The binding structure: a bipartite revisit (#416)

The proton was originally placed on a trivalent junction because the color singlet ε_{ijk} is absent from $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and first appears in $\mathbf{3}^{\otimes 3}$. That theorem rules out a *generic* bipartite merge, but not two bipartite steps with a projection: if the first merge projects onto the antisymmetric diquark $\bar{\mathbf{3}} = \varepsilon qq$, then $\bar{\mathbf{3}} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8}$ contains the singlet. The prism extrusion already accepted a cumulative *twist* permutation, an orientation-reversing tube that is a geometric antisymmetrizer; #416 exposed it cleanly as `RegisterTopology::orientationReversingTwist / setTwist` with `Cobordism::twistedCylinder`.

Two results landed. First, the twisted bipartite tube projects the emergent diquark onto $\bar{\mathbf{3}}$ (the antisymmetric channel) rather than the generic symmetric average — the orientation-reversing twist *is* the geometric antisymmetrizer the representation theory wanted. Second, and more consequential:

Finding 3 (Confinement is not fundamental in a connected bulk). The deeper obstruction was that the colored diquark intermediate is non-carriable (confinement): only $\sigma = 0$ combinations carry as *free boundary* states. A *connected* two-body path geometry — where the diquark is internal and never a free boundary — evades this. Measured, the connected path reaches the singlet with reachability 1.00, versus 0.55 for the welded variant. The colored intermediate’s non-carriability is therefore *not* a fundamental wall: it is hosted in the bulk of one connected relaxation. This single fact reshapes the assembly architecture (§9).

6 The flavor sector: a no-go (#414)

Proton (uud) and neutron (udd) differ only by flavor; the color-only geometry cannot tell them apart. Principle 1 sought flavor as a split of the existing register. That hope is now *refuted*, cleanly and for a structural reason.

Principle 2 (Flavor cannot share the color register). *Any relabeling-invariant per-window measurable commutes with the window-cycling automorphism g that realizes the color $\mathbb{Z}_3$, hence is constant on g 's orbit. The uud/udd discriminator is a function on that orbit, so it collapses: it cannot distinguish windows that g identifies. Measured, the discriminator margin is $\sim 3 \times 10^{-6}$, bounded by the intertwining residual itself — i.e. machine zero, no signal.*

The reason is symmetry parity. The color singlet is the symmetry-*even* ($A_4/\mathbb{Z}_3$ invariant) line; flavor is symmetry-*odd* (it must *distinguish* the windows the symmetry identifies). A single register cannot host both an even invariant and an odd discriminator on the same orbit. Neither candidate mechanism survives: the spatio-temporal split is relabeling-invariant (so it is constant on the orbit), and the Dirac–Kähler taste multiplicity is an operator-sector degeneracy, not a per-window state label.

Finding 4 (The fix is symmetry-breaking isospin, not a dimensional artifact). The collapse is *not* an artifact of working in $2 + 1$ D or of the discretization — it is forced by the equivariance of the read-out. The cure is a symmetry-*breaking* isospin doublet: a genuinely new construction in which the u/d asymmetry is put in as a broken $SU(2)$ that *acts* (the ladder $p \leftrightarrow n$), exactly as the color $\mathbb{Z}_3$ acts through P_{out} . A static even register cannot be coaxed into producing an odd label; one must break the symmetry on purpose. This is the track's sharpest negative result and it redirects the flavor follow-up away from “find it in the existing register” toward “build the broken-isospin doublet.”

7 The dimensional question (#418, #429)

Full $\vec{E}, \vec{B}$ and the full 4×4 Dirac algebra live only in $3 + 1$ dimensions — a genuine S^3 spatial slice extruded in time. The present junction is $2 + 1$ D ($S^2 \times I$), a reduced sector. #418 is the spike that scopes the $3 + 1$ D home and returns a go/no-go for the full electroweak content; #429 makes the apex interior dimension-generic so the spike has a buildable interior.

7.1 Dimension-generic apex interior and an $n \geq 4$ manifold check (#429)

`Spacetime::symmetricStackCells` now carries an `nApexSlices` parameter (default 1, reproducing the single-reflection #413 interior): each apex is a centre of point-reflection, its neighbour edges inverted through it to form the next slice plus a cap, stacked `nApexSlices` times. The interior connectivity generalizes by *coface-mirroring* — cone each top d -simplex to an apex and reflect apex-connectivity across a $(d - 1)$ -facet shared by two cofaces. The dimension-genericity is backed by a rigorous `ChainComplex::dualComplexIsValid` that extends to $n \geq 4$; the construction is *validated as a 4-manifold* over a tetrahedral S^3 base.

Finding 5 (The emergent twist is PT-symmetric, localized to creation/annihilation). Each apex is a -1 point-reflection, i.e. a PT inversion. Composed over the stack, the reflect-and-cap iteration tracks the γ^5 chiral split (an $8/8$ partition of the 16 form components) and carries the net $\mathbb{Z}_2$ spinor sign ($2\pi \mapsto -1$): the geometry realizes the spinor double cover, not merely the orientation

$\mathbb{Z}_2$. *Corrected reading, important:* the naive per-slice alternation $(-1)^j$ **over-reads** the structure. For a *freely propagating* worldline the apex cap is manifold-closure only — it carries no physical time reversal. The genuine PT / time-orientation reversal is *localized to creation/annihilation vertices*: an antiparticle is a particle running backward in time (Stückelberg–Feynman), so the PT flip belongs at the pair-creation/annihilation event, not at every slice of a propagating line. Reading a twist on every slice would manufacture spurious sign structure along free propagation; the physical content is concentrated at the events.

8 CI hygiene (#437)

Making the symmetric apex the default (#413/#429) and renaming the orientation class (#412) left five post-merge test failures. These were *stale prism-era tests*, not #429 regressions: they asserted the prism’s behavior in places where the default geometry had moved underneath them. The golden invariants G_1 – G_5 stayed green throughout. The fixes pin the tests to the right path: the photon test now exercises the *prism* path, where the spontaneous photon is real (Finding 1); the residual test asserts the machine-zero floor of the symmetric default; and the relabeling test pins the prism stride. The track’s shared regression module (`tests/cobordism/test_epic410_invariants.py`) remains the gate every PR must keep green.

9 Architecture pivot: bipartite junctions and one global relaxation

The track’s binding result (Finding 3) forces a change in how the proton is *assembled*. The tripartite W_{ABC} (`TripartiteRegisterTopology`) is **dropped** from the assembly path. The proton is built from *sequences of bipartite (two-body) junctions* relaxed in *one global step*, with color-emergent creation.

Why both at once. The colored diquark intermediate is floored by confinement if it is *pinned* as a free boundary state (a staged, separate-relaxation assembly fails exactly there). But it is *hosted* in the bulk of one connected relaxation — this is Finding 3’s reachability 1.00. So bipartite structure and global-relaxation are required *together*: pin only the initial and final states, relax the entire interior at once, and let the diquark and other intermediates *emerge*. Staging is at most a warm start, never a separate relaxation.

Charge $\otimes$ color. The pivot keeps the two sectors in their natural homes. Charge stays in the emergent temporal sector (the Gauss-law holonomy of §3.2, Finding 2); color stays in the register kernel. The assembly composes them without a parallel charge register — charge is read emergently from the relaxed Lorentzian geometry, not carried as a hand-weighted covector.

Creation, with the old assumptions quarantined. The assembly reuses only the *creation event* of the existing `pairCreate` primitive. The Charged-Cartan assumptions that surrounded it — a parallel $\hat{Q}$ vertex register, an imposed pair-Hamiltonian, Metropolis Monte-Carlo dynamics, a hand-dialed CP phase — are **quarantined**: they are not part of the emergent path. Creation is color-emergent (color-indefinite at birth, crystallizing from context) and charge is read by Gauss’s law.

10 Faithfulness: the golden invariants held

Every result above landed under a fixed set of proton invariants, the track’s G_1 – G_8 , shared in one regression module so later subtasks could not silently regress earlier ones.

	invariant	status across the epic
G_1	proton present	singlet overlap $\rightarrow 1.000000$ on the symmetric default (#413)
G_2	confinement	$\sigma \rightarrow 0$ for the symmetric input
G_3	charge conservation	$\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$ exact ($\sim 10^{-14}$)
G_4	topology	$b_1 = 11$ and a valid manifold; <code>dualComplexIsValid</code> now $n \geq 4$ (#429)
G_5	color $\mathbb{Z}_3$ intact	P_{out} eigenvalues $\{1, \omega, \omega^2\}$
G_6	relabeling-invariance	owned by #412; the no-go of #414 is a <i>consequence</i> of it
G_7	determinism	fixed seed $\Rightarrow$ reproducible; no wall-clock/random seeding
G_8	emergent-first	charge (#411), diquark (#416), twist (#429) all read off the relaxed geometry

The geometry change (#413) preserved the color sector exactly while sharpening the singlet; the orientation change (#412) preserved the manifold and spectrum while making the charge a label-independent invariant; the Dirac–Kähler and E/B additions (#415, #417) are observable-only. No golden result drifted. Notably, G_6 is not merely a guard here — the flavor no-go (§6) is *derived* from it: the same relabeling-invariance that makes the charge faithful is what forbids flavor from hiding in the existing register.

11 What is established, and what is open

Established (merged and measured). The geometry is honest (no label-sort artifacts, #412); the proton color singlet is *exact* (overlap 1.000000, residual 4.5×10^{-14} , #413); the field strength carries a faithful E/B split (#417); electric charge is an emergent, metric-robust Gauss-law holonomy that vanishes without Lorentzian worldlines (#411); the Dirac–Kähler operator gives exact Clifford structure, γ^5 chirality, a conserved current, and the 4×4 taste multiplicity (#415); a twisted bipartite tube projects the antisymmetric diquark and a connected path reaches the singlet at reachability 1.00 (#416); flavor as a register split is *ruled out* with the cure identified (#414); and the apex interior is dimension-generic, 4-manifold validated over S^3 , with the emergent twist correctly localized to creation/annihilation events (#418, #429). CI is green and pinned (#437).

Open work (named here, not created).

- **#435 — bipartite color-emergent creation primitive + charge $\leftrightarrow$ color bridge.** The foundation of the assembly pivot: a two-body $q/\bar{q}$ creation event, color-indefinite at birth, with charge read by Gauss’s law and the Charged-Cartan assumptions quarantined. Blocks #434 and #438.
- **#434 — Experiment A: emergent intermediates.** The whole $q/\bar{q} \rightarrow$ proton & anti-proton event as one connected bipartite cobordism; pin only initial and final, relax the entire interior at once, let the diquark and intermediates emerge; watch for sourced photons.
- **#438 — Experiment B: fixed bipartite sequence + A-vs-B comparison.** Reuse #434’s structure but pin the known intermediates ($q + q \rightarrow \bar{\mathbf{3}}$ diquark, diquark + $q \rightarrow$ proton, and conjugates); confirm the colored diquark is hosted in the bulk (low r_U , per #416) and compare the emergent path A against the imposed path B.

- **Flavor: the symmetry-breaking isospin doublet.** The #414 cure — a broken $SU(2)$ that *acts* ($p \leftrightarrow n$ ladder), a genuinely new construction, not a split of the existing register.
- **#429's equivariant $d \geq 3$ gap-fill.** Close the remaining coface-mirroring generalization so the dimension-generic apex is fully equivariant in $d \geq 3$, completing the $3 + 1$ D interior the #418 spike needs.
- **Flux-tube creation energetics.** The energy cost of color-emergent creation and the flux-tube structure between the bipartite vertices — the dynamical content the assembly experiments will probe.

Verdict. The track delivered its geometric foundations and three of its four sectors as merged, machine-verified results; it killed two hypotheses cleanly (flavor in the existing register; the per-slice twist reading) and answered the binding question (confinement is not fundamental in a connected bulk). The net is a sharper architecture — bipartite junctions, one global relaxation, color-emergent creation, charge in the temporal sector and color in the register — and a precise list of what to build next. As throughout the program: we read what the geometry produces, we do not impose the answer, and the falsifiable tests of §10 arbitrate.